

# ASWQM

## Autonomous Sail Boat for Water Quality Monitoring

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### Abstract

Polluted water poses health risks and limits traditional water monitoring techniques because of potential dangers on workers. So We developed a remotely operated boat equipped with sensors to monitor contaminated water safely. The boat accesses polluted areas using a wireless communication protocol , allowing workers to remain at a safe distance. Sensors on the boat measure parameters to provide continuous real-time data on water conditions.

### Background/Intro

Traditionally, water quality monitoring relies on manual systems of data collection sampling and sending it to a laboratory for analysis. Monitoring water quality in remote locations can be challenging due to the need for frequent manual sampling and testing. This relies on human intervention which can be inconvenient, time-consuming and expensive. An autonomous water quality monitoring system can provide a solution by continuously monitoring the water without requiring physical access by using mobile application

### Team Members

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### Objectives

- Accurate and up-to-date data on water conditions
- Monitor remote locations that are difficult to access frequently for manual sampling
- Reduce the costs of water quality testing by eliminating the need for frequent human sampling and lab tests
- Automatically classify the water as either "drinkable" or "polluted" based on the sensor readings and pre-defined thresholds

### Methods

#### Water Sensors

- PH Sensor
- Turbidity Sensor
- TDS
- Water Temperature Sensor

#### Mobile Application

serves as the interface for controlling the boat and displaying sensor data.

#### Motion

User send new location, MPU adjust the direction to target location and the boat start its mission

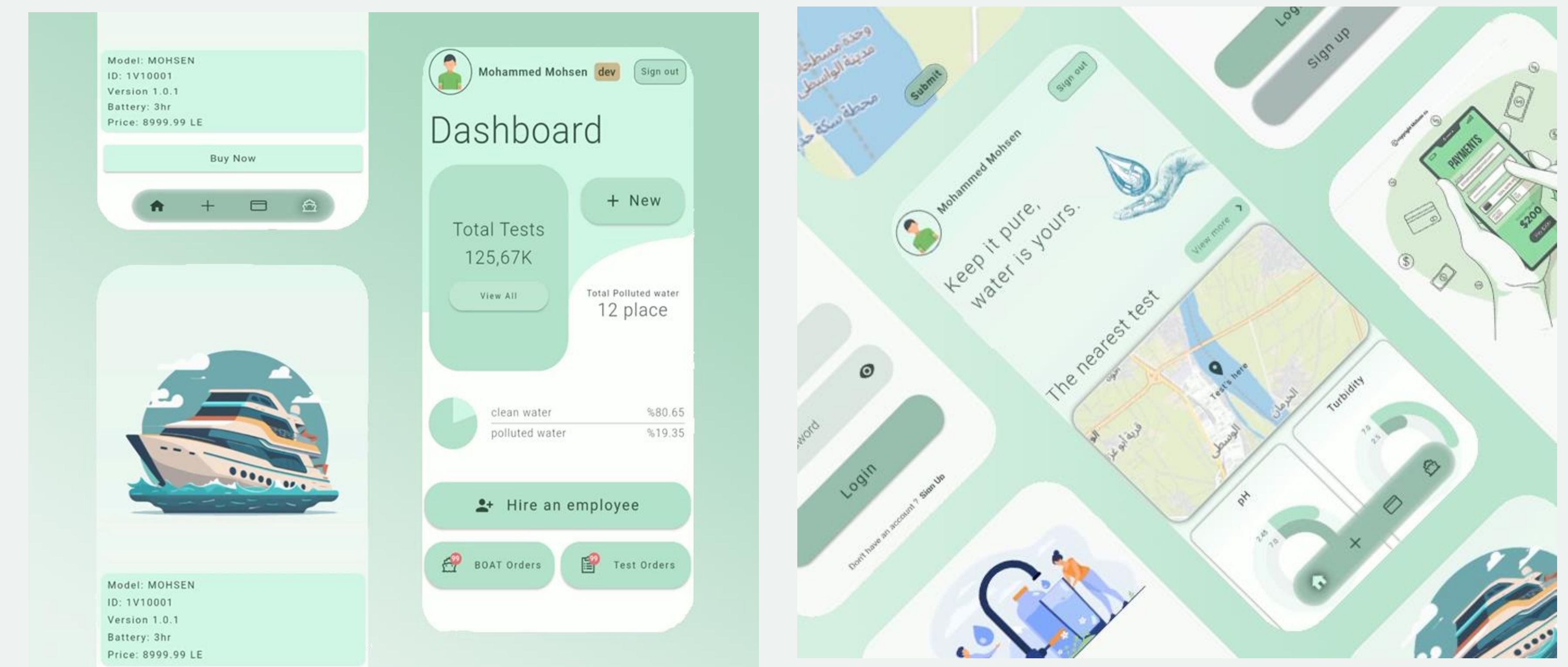
### Results

#### Water Quality Parameters Measured

The sensors successfully measured pH from 6.5 to 8.5 , turbidity from 0 to 120 NTU, dissolved particles from 0 to 20 mg/L, and temperature from 20 to 30 °C. After a rainstorm, the system detected a rapid rise in turbidity from 10 to 90 NTU and a drop in pH from 7.8 to 6.8.

#### Effectiveness of Mobile Application

The application allowed intuitive control of the boat's movement and display sensor data). Personnel were able to operate the system with little training.



#### Ability to Navigate to Desired Locations

The boat successfully navigated to target locations within 5 meters of the specified coordinates on 90% of missions.

Improved MPU

calibration reduced directional errors that initially caused some failure to reach targets.

